

Week: 10
Experiment: 09

“Estimation of 2 Different Types of Chlorophyll Present in Leaves Using a Spectrophotometer”

1) Lab Performance (10 Marks):

2) Total Marks (20 Marks):

3) Observation (5 marks):

Plant Name: Pisonia

		Wavelength	Optical Density/Absorbance
Sunlight	Chlorophyll B	645	0.422
	Chlorophyll A	663	1.029
Dark	Chlorophyll B	645	0.338
	Chlorophyll A	663	0.819

Chlorophyll A showed higher absorbance than Chlorophyll B in both sunlight and dark conditions. Leaves exposed to sunlight had greater optical density for both

pigments compared to leaves kept in the dark. Chlorophyll A exhibited maximum absorbance at 663 nm (red light), while Chlorophyll B peaked at 645 nm (blue light).

The higher absorbance in sunlight-exposed leaves indicates a greater chlorophyll concentration due to active photosynthesis, whereas the lower absorbance in dark-kept leaves reflects reduced chlorophyll formation.

4) Calculations (10 marks): (Only do the Calculation for your leaves sample, e.g., either dark or light)

Formula for Chlorophyll a:

- **Chlorophyll a (mg/mL) = $12.7 \times A_{663} - 2.69 \times A_{645}$**
Total Chlorophyll a (mg) = Chlorophyll a (mg/mL) $\times$ Final volume (mL)
= $11.93 \times 30 = 357.99\text{mg}$

Formula for Chlorophyll b:

- **Chlorophyll b (mg/mL) = $22.9 \times A_{645} - 4.68 \times A_{663}$**
- **= $22.9(0.422) - 4.68(1.029) = 4.85 \text{ mg/ml}$**
Total Chlorophyll b (mg) = Chlorophyll b (mg/mL) $\times$ Final volume (mL)
= $4.85 \times 30 = 145.44\text{mg}$

Formula for Total Chlorophyll:

- **Total Chlorophyll (mg/mL) = Chlorophyll a + Chlorophyll b**
- **= $11.93 + 4.85 = 16.78\text{mg/ml}$**
Total Chlorophyll (mg) = Total Chlorophyll (mg/mL) $\times$ Final volume (mL)
 $16.78 \times 30 = 503.4\text{mg}$

5) Result (5 marks): (Should include both dark and sunlight samples)

Leaves exposed to sunlight had a higher total chlorophyll concentration (16.78 mg/mL) compared to leaves kept in the dark (13.40 mg/mL), showing that light plays a crucial role in chlorophyll production. In both conditions, Chlorophyll A was more abundant than Chlorophyll B. Specifically, in sunlight, Chlorophyll A measured 11.93 mg/mL, while in the dark it was 9.49 mg/mL.

The total chlorophyll extracted from the sunlight-exposed leaves was 503.4 mg, whereas the dark-kept leaves yielded only 402.0 mg, a decrease of over 100 mg. Chlorophyll B also decreased from 4.85 mg/mL in sunlight to 3.91 mg/mL in the dark, further emphasizing that chlorophyll formation relies on light exposure.

Despite the differences in concentration, the ratio of Chlorophyll A to Chlorophyll B remained relatively constant 2.46 in sunlight and 2.43 in the dark indicating that the proportion between the two pigments does not change significantly with light conditions.

6) Problems:

May be the pigment was not fully extracted from the leaf so I may have caused underestimation of total chlorophyll content present in the leaf.